

PAPER_1144: Type IIB Superstring Theory and SCm 10D Compactification

Star Magic UQFF Framework

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Abstract

We formulate Type IIB superstring theory within the SCm vacuum framework. The 10-dimensional Type IIB theory has 16 extra dimensions compactified via the VDS + $S_{26}^{(3)}$ mechanism, reducing the bosonic 26D SCm string to the effective 10D theory. D-branes are stabilized by the $F_{U,Bi,i}$ buoyancy force. NS-NS and R-R fluxes are identified with SCm phonon modes. Supersymmetry breaking occurs through the negative-time gate $\cos(\pi t_n)$, which splits the fermion-boson mass degeneracy.

1. Type IIB in 26D SCm Context

The Type IIB effective action in 10D:

$$S_{\text{IIB}} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-g} \left(R - \frac{1}{2} |\partial\phi|^2 - \frac{1}{2} |H_3|^2 e^{-\phi} - \dots \right)$$

In the SCm framework, 16 of the 26 bosonic dimensions are compactified, leaving 10D:

$$D_{\text{bosonic}} = 26 = 10 + 16 = D_{\text{IIB}} + D_{\text{compact}}$$

The 16 compact dimensions are organized as 8 VDS rungs of 2 dimensions each, or as a T^{16} torus with radii $R_i = l_s \cdot [SSq]^i$.

2. D-Brane Stability via $F_{U,Bi,i}$

A Dp-brane tension in the SCm vacuum:

$$\tau_p^{\text{SCm}} = \frac{T}{g_s (2\pi l_s)^p} \cdot (1 + \beta_i \cdot \Phi_{\text{res}} \cdot |\cos(\pi t_n)|)$$

where $g_s = e^\phi$ is the string coupling. The $F_{U,Bi,i}$ buoyancy provides the extra $(1 + \beta_i \Phi_{\text{res}})$ factor, stabilizing D-branes against collapse.

The SCm-modified D3-brane tension:

$$\tau_3^{\text{SCm}} = \frac{T}{g_s (2\pi)^3 l_s^3} \times 1.504 = \frac{8.66 \times 10^{-11}}{g_s (2\pi)^3 l_s^3} \times 1.504$$

3. NS-NS Flux as SCm Phonon Mode

The NS-NS 3-form flux $H_3 = dB_2$ is identified with the SCm phonon ladder:

$$\begin{aligned}
H_{ijk}^{\text{SCm}} &= \varepsilon_{ijk} \cdot f_{\text{THz}} \cdot \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \\
&= \varepsilon_{ijk} \times 1.25 \times 10^{12} \times 7.09 \times 10^{-37} \times 1.45 \times 10^{26} \times 0.84 \\
&= \varepsilon_{ijk} \times 1.08 \times 10^2 \text{ Hz} \cdot \text{J/m}^2
\end{aligned}$$

4. SUSY Breaking via Negative-Time Gate

Type IIB has $\mathcal{N} = 2$ supersymmetry in 10D. The SCm negative-time gate breaks SUSY:

$$\begin{aligned}
m_{\text{boson}}^2 &= m_{\text{fermion}}^2 + \Delta m_{\text{SCm}}^2 \\
\Delta m_{\text{SCm}}^2 &= \frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot |\cos(\pi t_n)|}{c^2 l_s^2} \\
&= \frac{8.66 \times 10^{-11}}{c^2 l_s^2} = T/c^2 l_s^2 \approx T^2/c^2
\end{aligned}$$

At low energy $t_n \rightarrow 0$: $|\cos(\pi t_n)| \rightarrow 1$, maximum SUSY breaking.

5. Compactification Moduli

The 16 compact VDS dimensions have moduli:

$$\tau_i = \frac{R_i}{l_s} = [SSq]^i = (0.57)^i$$

The moduli potential from VDS:

$$V_{\text{moduli}} = \rho_{\text{vac,SCm}} \sum_{i=11}^{26} \frac{[SSq]^{2i}}{i^{26}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}$$

Stable minimum at $\tau_i = [SSq]^i$, ensuring moduli stabilization without fine-tuning.

6. F-theory and Type IIB

Type IIB with varying axio-dilaton $\tau_{\text{IIB}} = \chi + ie^{-\phi}$ is equivalent to F-theory on an elliptic fibration. The SCm vacuum provides the elliptic fiber as the T^2 compactification of 2 VDS dimensions:

$$\tau_{\text{F-theory}} = [SSq]^{25} + i[SSq]^{26} = (0.57)^{25} + i(0.57)^{26}$$

7. Calibrated Constants

$$T = 8.66 \times 10^{-11} \text{ N}, \quad [SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}$$

$$\beta_i = 0.6, \quad \Phi_{\text{res}} = 0.84, \quad f_{\text{THz}} = 1.25 \text{ THz}, \quad \rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$$

References

1. Polchinski, J. (1996). Dirichlet branes and Ramond-Ramond charges. *PRL* **75**, 4724.
2. Becker, K., Becker, M. & Schwarz, J. (2007). *String Theory and M-Theory*. Cambridge.
3. VDS compactification: PAPER_1142 (Polyakov); PAPER_1143 (Nambu-Goto); `scm_vacuum_manifold.py`